

EE3621: Digital Signal Processing

Lecture 12: IIR Filter Design — Analog Prototype & Bilinear Transform (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins): IIR Design Philosophy** — Borrowing from continuous-time analog filter theory.
 - **05:00 – 15:00 (10 mins): Analog Prototypes** — Butterworth, Chebyshev (I & II), and Elliptic filters.
 - **15:00 – 20:00 (5 mins): Impulse Invariance Method** — Time-domain mapping and aliasing.
 - **20:00 – 28:00 (8 mins): The Bilinear Transform** — Frequency-domain mapping and warping.
 - **28:00 – 35:00 (7 mins): Bilinear Design Procedure** — Worked example for a 3rd-order Butterworth.
 - **35:00 – 40:00 (5 mins): Checkpoints** — Review questions.
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2 IIR Design Philosophy

Infinite Impulse Response (IIR) digital filters are typically designed by first designing a continuous-time analog prototype. The standard approach involves:

1. **Pre-warping:** Convert digital frequency specs to analog frequency specs.
2. **Analog Design:** Compute the analog transfer function $H_a(s)$.
3. **Transformation:** Map $H_a(s)$ to the digital transfer function $H(z)$.

3 Butterworth Analog Prototype

The Butterworth filter is maximally flat in the passband.

3.1 Magnitude Response

The squared magnitude response is:

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}} \quad (1)$$

3.2 Derivation of Pole Locations

Substituting $s = j\Omega$ ($\Omega = s/j$):

$$H_a(s)H_a(-s) = \frac{1}{1 + \left(\frac{s/j}{\Omega_c}\right)^{2N}} = \frac{1}{1 + (-1)^N \left(\frac{s}{\Omega_c}\right)^{2N}} \quad (2)$$

Setting the denominator to zero:

$$1 + (-1)^N \left(\frac{s}{\Omega_c}\right)^{2N} = 0 \implies \left(\frac{s}{\Omega_c}\right)^{2N} = (-1)^{N-1} \quad (3)$$

The stable poles (LHP) are:

$$s_k = \Omega_c e^{j\frac{\pi}{2N}(2k+N-1)}, \quad \text{for } k = 1, 2, \dots, N \quad (4)$$

These poles lie precisely on a circle of radius Ω_c .

3.3 Derivation of Order Formula

Given passband attenuation A_p at Ω_p and stopband attenuation A_s at Ω_s :

$$\left(\frac{\Omega_p}{\Omega_c}\right)^{2N} = A_p^2 - 1 \quad \text{and} \quad \left(\frac{\Omega_s}{\Omega_c}\right)^{2N} = A_s^2 - 1 \quad (5)$$

Dividing and solving for N :

$$N \geq \frac{\log_{10} \left(\frac{A_s^2 - 1}{A_p^2 - 1} \right)}{2 \log_{10} \left(\frac{\Omega_s}{\Omega_p} \right)} \quad (6)$$

4 Chebyshev and Elliptic Prototypes

- **Chebyshev Type I:** Equiripple passband, monotonic stopband. $|H(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 C_N^2(\Omega/\Omega_p)}$. Poles lie on an ellipse.
- **Chebyshev Type II:** Monotonic passband, equiripple stopband.
- **Elliptic (Cauer):** Equiripple in both bands, providing the minimum possible order. Uses Jacobian elliptic functions.

Filter Type	Passband	Stopband	Phase Linearity	Order Required
Butterworth	Maximally Flat	Monotonic	Best	Highest
Chebyshev I	Equiripple	Monotonic	Moderate	Lower
Chebyshev II	Monotonic	Equiripple	Moderate	Lower
Elliptic	Equiripple	Equiripple	Non-linear	Lowest

Table 1: Comparison of Analog Prototype Filters

5 The Impulse Invariance Method

The impulse invariance method defines $h[n] = T_s h_a(nT_s)$. An analog pole at s_k maps directly to a digital pole at $z_k = e^{s_k T_s}$:

$$H_a(s) = \sum_{k=1}^N \frac{A_k}{s - s_k} \implies H(z) = \sum_{k=1}^N \frac{T_s A_k}{1 - e^{s_k T_s} z^{-1}} \quad (7)$$

Drawback: Inherently suffers from aliasing, rendering it useless for highpass/bandstop filters.

6 The Bilinear Transform (BLT)

The BLT applies the conformal mapping:

$$s = \frac{2}{T_s} \frac{1 - z^{-1}}{1 + z^{-1}} = \frac{2}{T_s} \frac{z - 1}{z + 1} \quad (8)$$

This maps the entire $j\Omega$ axis onto the unit circle without aliasing. By substituting $s = j\Omega$ and $z = e^{j\omega}$, the frequency warping relationship is:

$$\Omega = \frac{2}{T_s} \tan\left(\frac{\omega}{2}\right) \quad (9)$$

7 Worked Example: 3rd-Order Digital Butterworth LPF

Design a 3rd-order digital Butterworth LPF with $\omega_c = 0.3\pi$ using BLT.

1. **Prewarping:** Let $T_s = 2$. $\Omega_c = \tan(0.15\pi) \approx 0.5095$ rad/s.

2. **Analog Prototype Design:** For $N = 3$, the poles are:

$$\begin{aligned} s_1 &= 0.5095e^{j120^\circ} = -0.2548 + j0.4412 \\ s_2 &= 0.5095e^{j180^\circ} = -0.5095 \\ s_3 &= 0.5095e^{j240^\circ} = -0.2548 - j0.4412 \end{aligned}$$

The transfer function is:

$$H_a(s) = \frac{\Omega_c^3}{(s - s_1)(s - s_2)(s - s_3)} = \frac{0.1323}{s^3 + 1.0191s^2 + 0.5193s + 0.1323} \quad (10)$$

3. **Bilinear Transform:** Substitute $s = \frac{z-1}{z+1}$:

$$H(z) = \frac{0.1323(z+1)^3}{(z-1)^3 + 1.0191(z-1)^2(z+1) + 0.5193(z-1)(z+1)^2 + 0.1323(z+1)^3} \quad (11)$$

Expanding and collecting terms yields:

$$H(z) = \frac{0.0495(1 + 3z^{-1} + 3z^{-2} + z^{-3})}{1 - 1.1618z^{-1} + 0.6959z^{-2} - 0.1378z^{-3}} \quad (12)$$

8 Checkpoints

1. **Q: Why use BLT for highpass filters instead of Impulse Invariance?**

A: Impulse Invariance causes spectral aliasing. Since highpass filters are not bandlimited, aliasing ruins the passband. BLT maps the whole axis without aliasing.

2. **Q: What happens to Butterworth poles as order N increases?**

A: The poles crowd along the circle of radius Ω_c and the ones near the $j\Omega$ axis create a steeper transition band roll-off.

3. **Q: Why choose $T_s = 2$ in BLT?**

A: It simplifies calculations; T_s scales the prewarped Ω_c and then is identically cancelled out during the $s \rightarrow z$ substitution, having no effect on final coefficients.